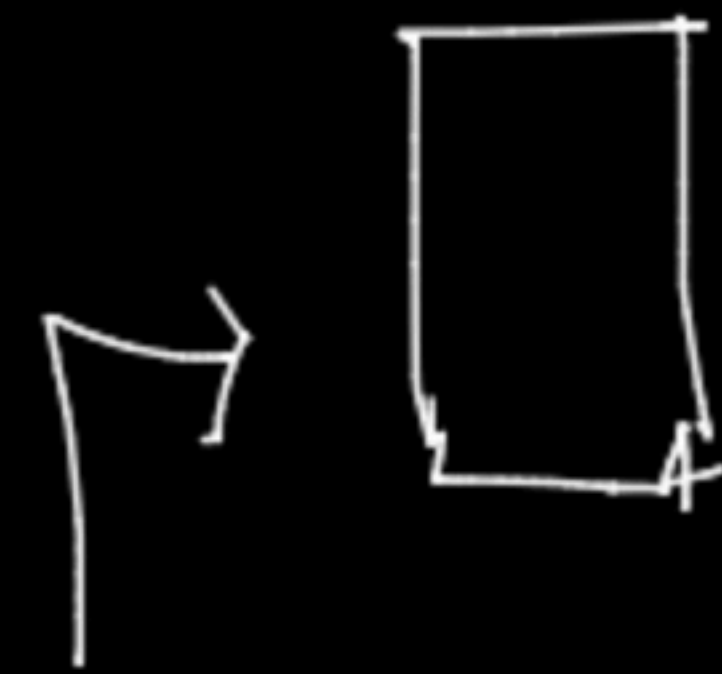


ML is data driven

Problem

10 feature 20 feature $\rightarrow$ 10 lakh



Setosa / Versa / W

Manual Rule $\rightarrow$ k

(Miss) Rule $\rightarrow$ 'hard'

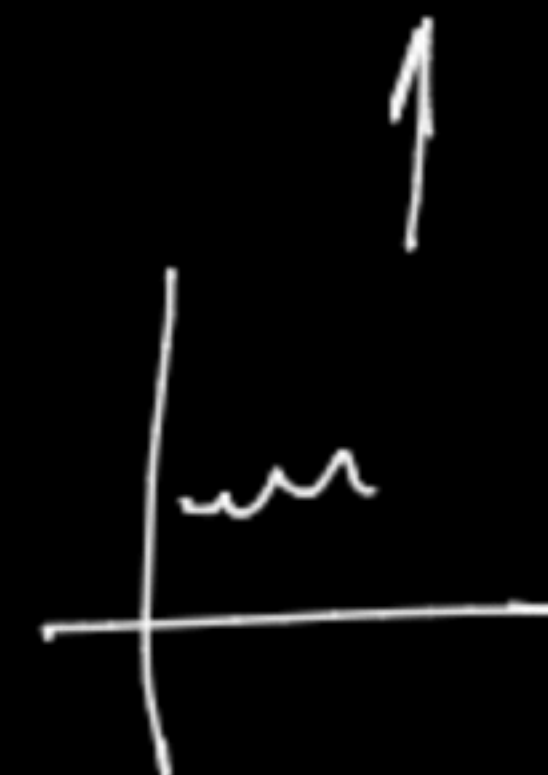
[SL, PL, PW, SW]

Rule

if

PLYS and SL $\rightarrow$ setosa

Analysis



time

'frob' y

system

Algorithms

ML

'mathematical / tech' 'Data'

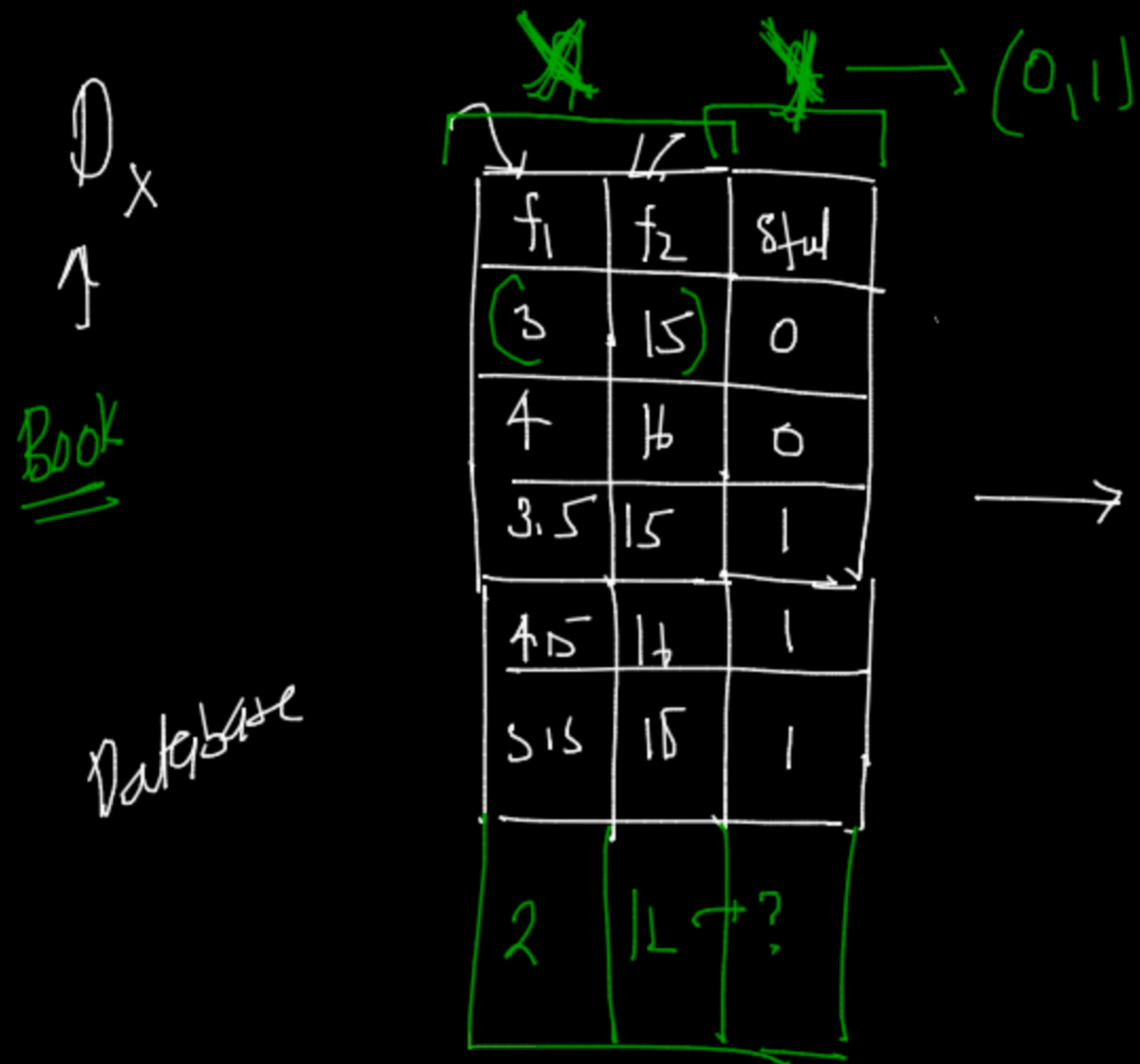


Rule

f()

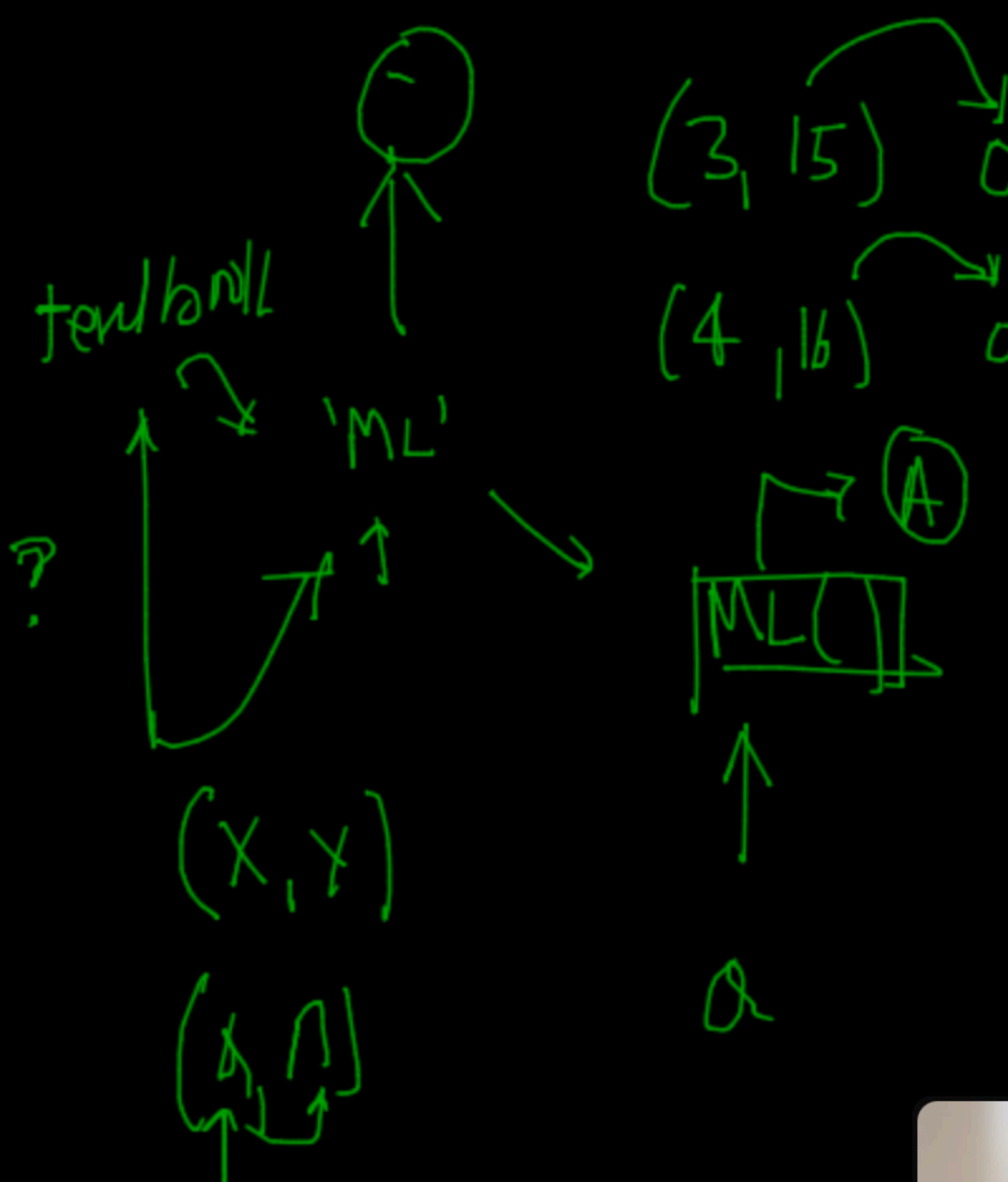
Classification
Multi-Class Classification



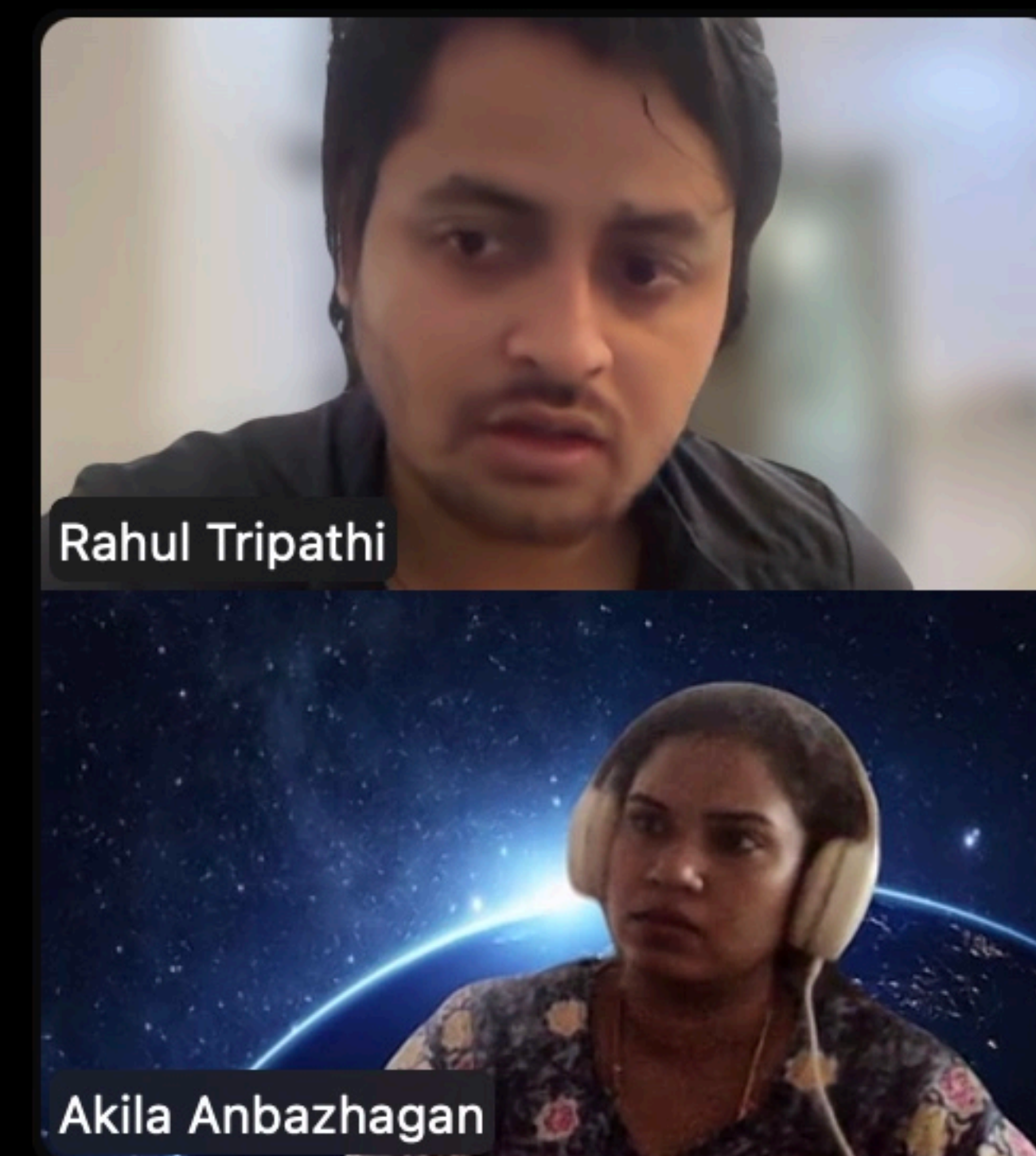
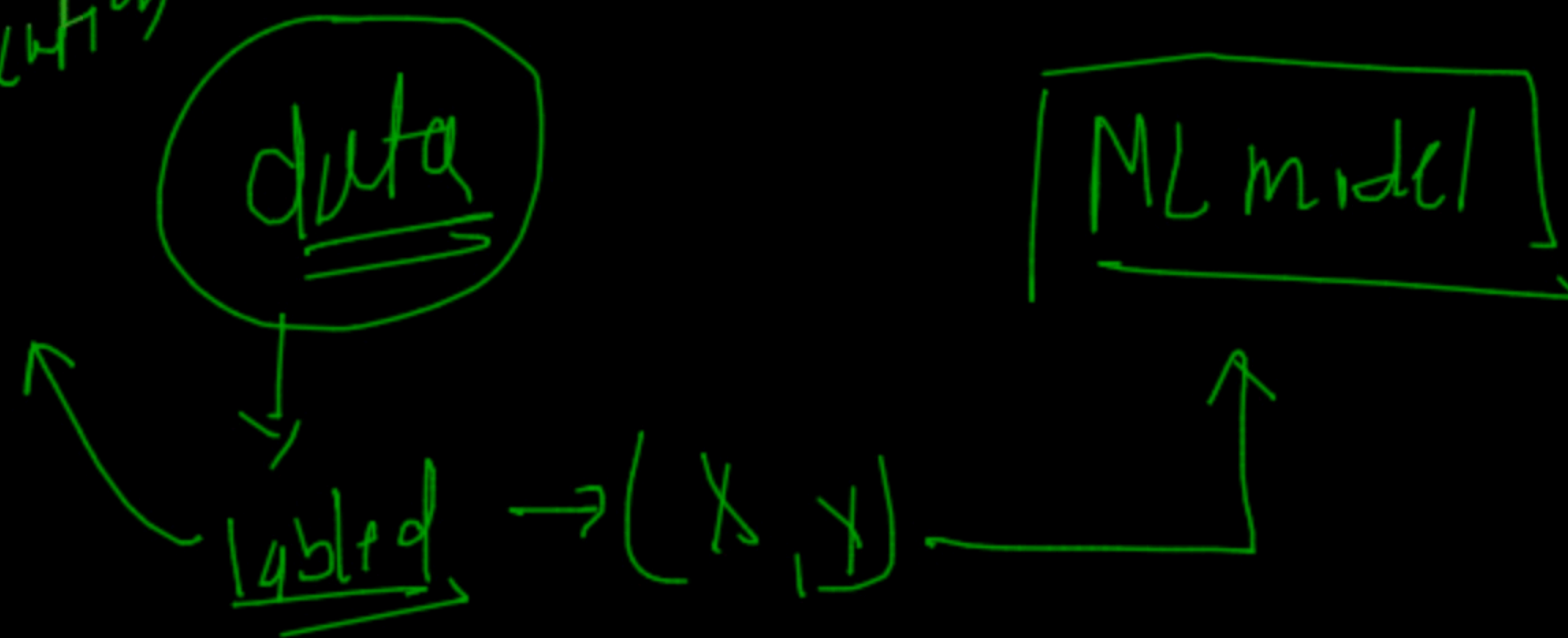


$[f_1, f_2]$
 $[2, 12]$

Yes



Binary classification



D_x

X		Y
Discount	Day	Sales
10%	T	450
8%	F	350
7%	F	300
⋮	⋮	⋮
⋮	⋮	⋮

Real

?

→ ?

$[15, T]$

(- -)

D_x

Data → $Y \in \text{Real}$

'Regression'



Rahul Tripathi



Akila Anbazhagan

way of training model

(Q, A)

10+

ML

Supervised ML

$P_X(x, y)$

python
↑
sci-
'sklearn'

3/1/ve

$Y \in [0, 1, \dots, K]$

$Y \in \text{Real}$

✓ 'Classification'

✓ Regression

- (i) KNN
- (ii) Log Regression

- (i) Linear Regression
- (ii) KNN

$P_X(x)$

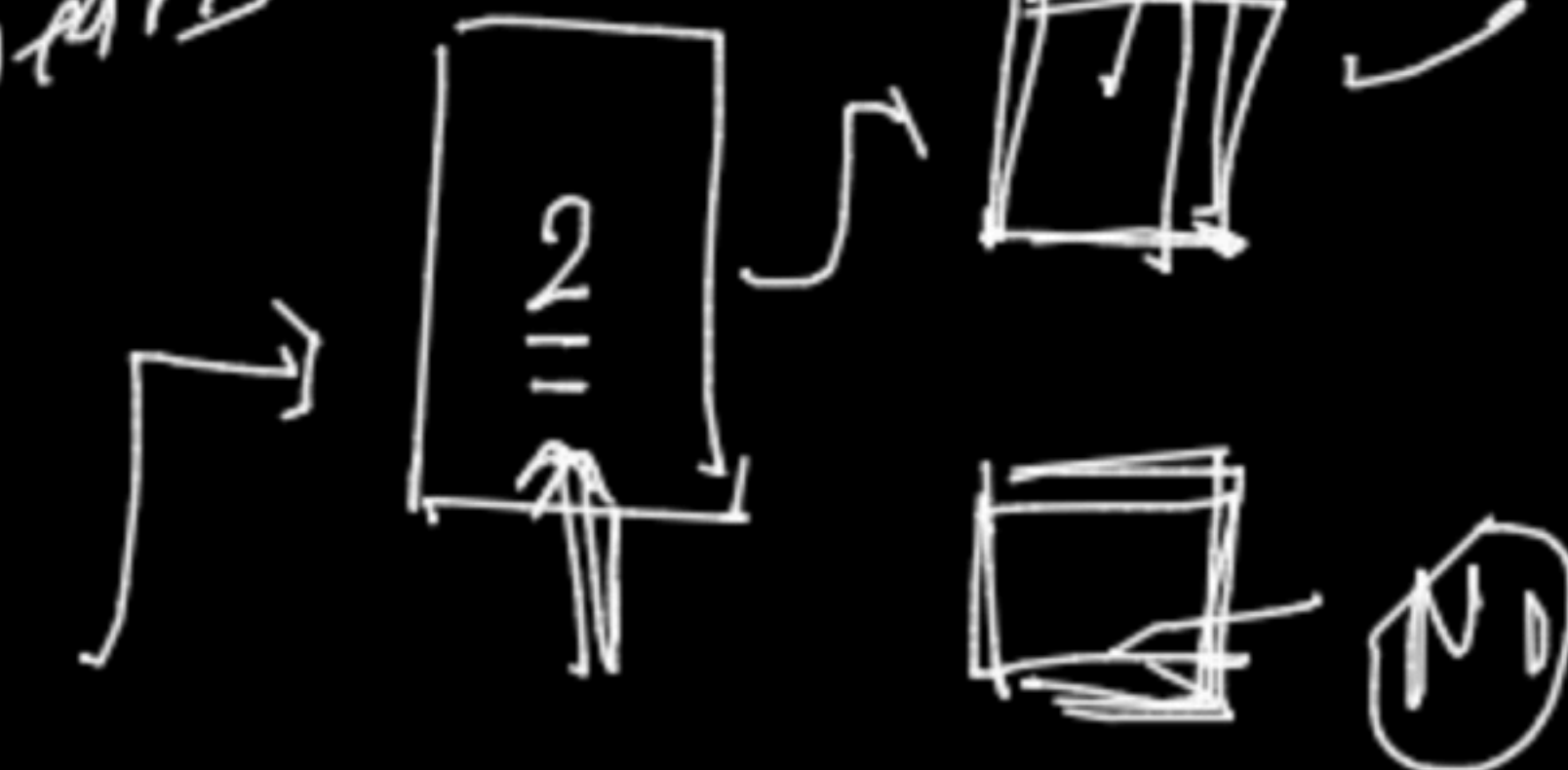
customer seg
C1
C

Unsupervised ML

$P_X(x)$

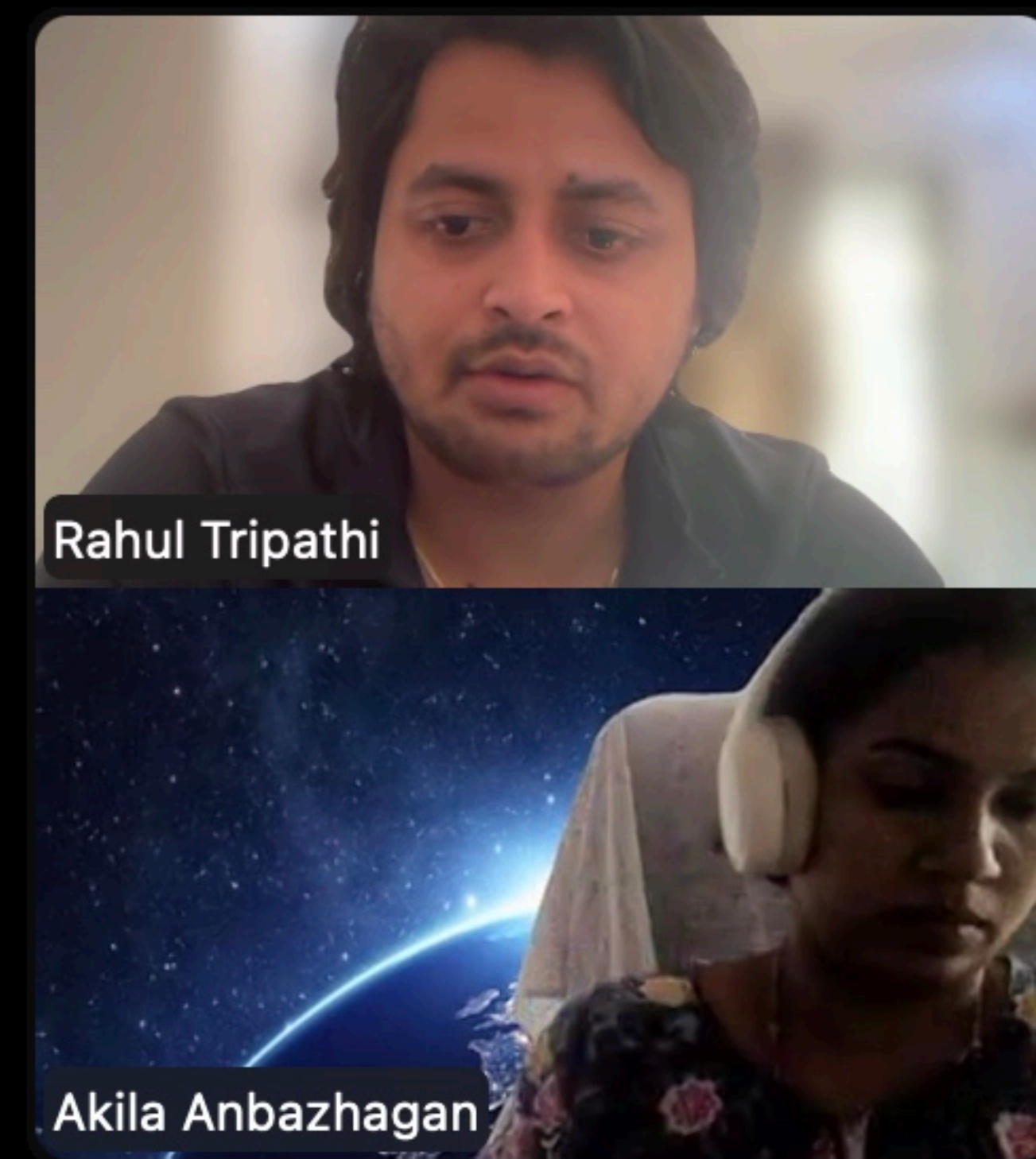
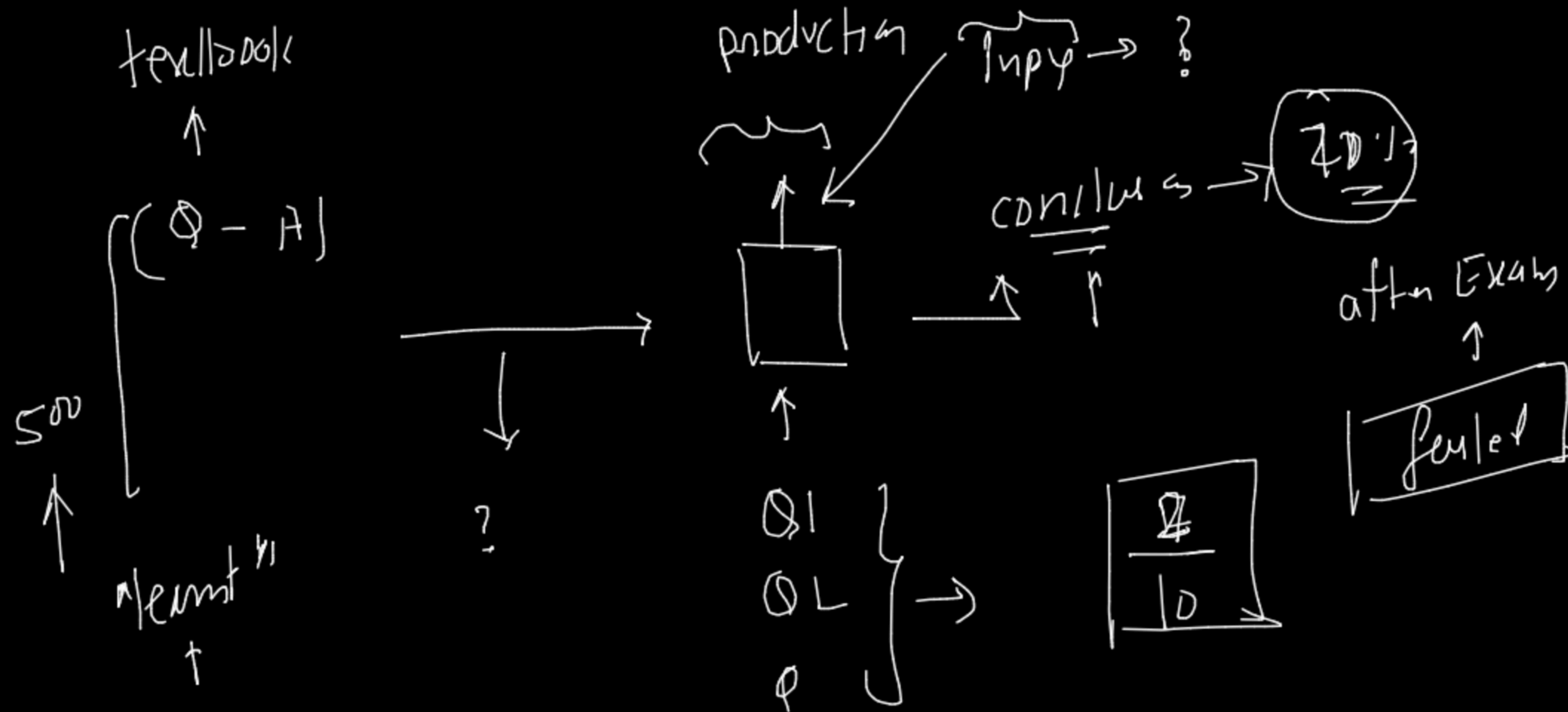
^(Clustering Algorithm)

(i) K-means



$P_X(x)$





$D_X(X, Y) \rightarrow D_{\text{Train}}(X_{\text{train}}^{80\%}, Y_{\text{train}}^{80\%}) \rightarrow \boxed{M} \rightarrow f()$

Random splitting

$\boxed{D_{\text{Test}}(X_{\text{test}}, Y_{\text{test}})}^{20\%}$

sklearn
train-test-split

x	y

KNF

1	0	1
1	1	1

$f()$

$f()$

x_1, x_2, x_3

model perform

x	y		
0	1	x	
0	0	✓	
1	1	✓	

$\frac{2}{3}$

$\Rightarrow 60\%$



KNN → Nearest Neighbours
 ↑
 math

$K \rightarrow \text{int}$

Assgn
 eg.
 =
 → `libsvm + sklearn`

small
 $K=3$
 1 → approve
 0 → reject

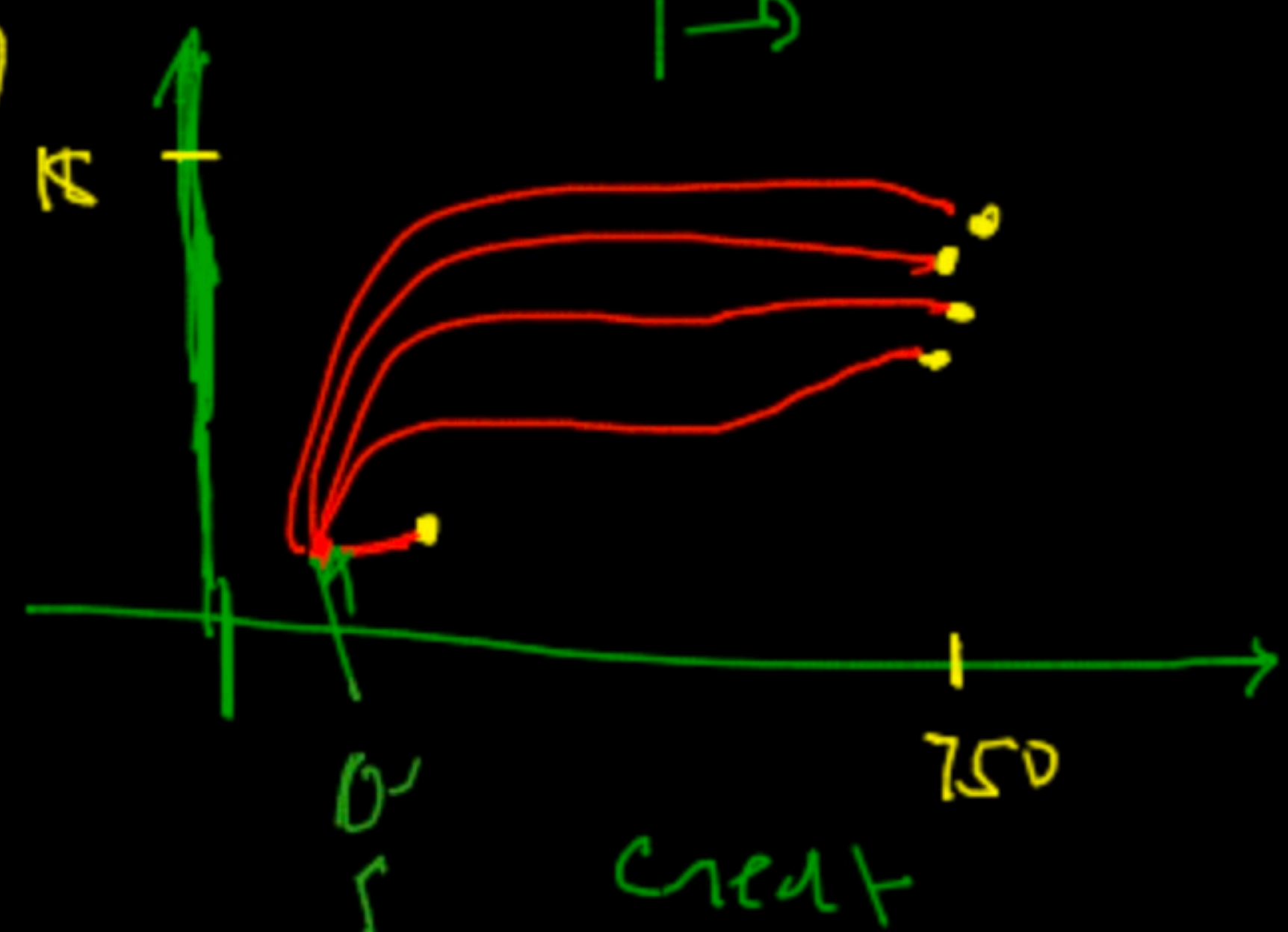
KNN
 ↑
 at time of prediction
 $K=$

↓
 distance

credit-score	income	status
(750	16)	1
(780	17)	1
(765	15)	1
(750	14)	1
(450	7)	0
(459	9	0
464	11	0

distance
 1.65 (2) ✓
 1.85 (3) ✓
 1.85 (4)
 1.86 (5)
 0.123 (1) ✓

vote
 2 → 1
 1 → 0
 → 1



(459, 9)



$K=?$
=

$D_{train}(X_{train}, Y_{train}) \rightarrow KNN(\text{5}, D_{train})$

~~7 9 13~~

$2 \rightarrow 0$
 $1 \rightarrow 1$

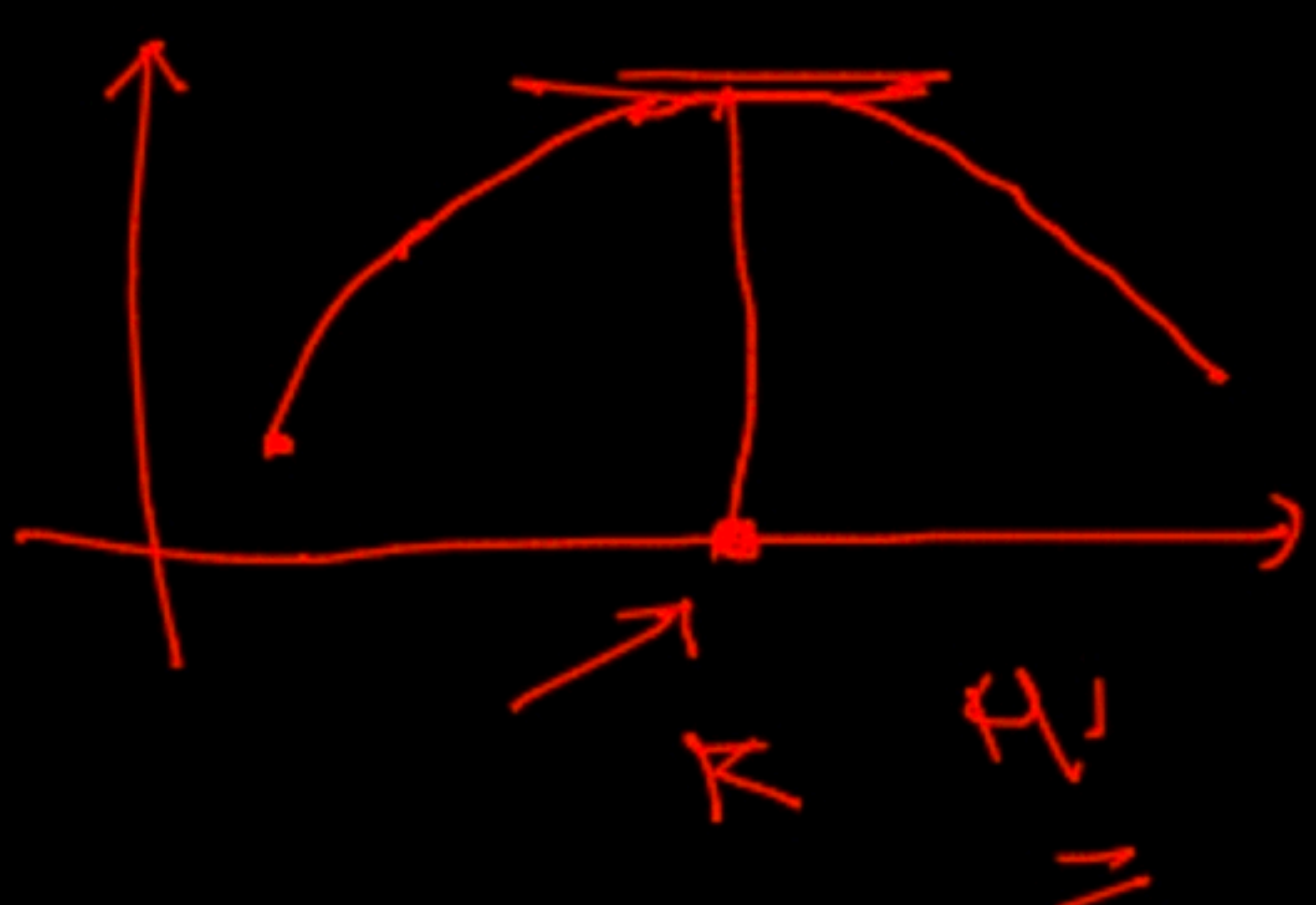
Y_{pred}, Y_{Actu}

x	y
1	1
2	0
3	0

2
3

X_{test}

76.1%
75.1%
78.1%
82.1%
78.1%



$K=9$ Best

